

Clinical Validation of a Deep Learning Model for Automated Detection of Cerebrovascular Steno-occlusive Disease on MRA

Purpose: The screening for intracranial aneurysms often overlooks concurrent steno-occlusive disease, which carries significant risk for ischemic events. This study sought to develop and validate a deep learning-driven solution for the automated identification of cerebrovascular stenosis and occlusion using MR Angiography to mitigate diagnostic oversight.

Methods: A robust neural network architecture was optimized using a large-scale, heterogeneous dataset of 1,722 TOF-MRA volumes, which included 1,366 stenotic segments and 524 total occlusions meticulously curated for high-fidelity training. The model's diagnostic performance was evaluated on an independent external validation (n=408), consisting of 165 confirmed pathological cases and 243 normative controls.

Results: In the validation analysis, the AI framework demonstrated superior screening performance with a sensitivity of 95.8% (correctly identifying 158/165 patients) and a specificity of 86.4% (210/243 controls). The system exhibited high-throughput efficiency, with a mean inferential latency of approximately 26.5 seconds per case, facilitating rapid clinical integration.

Conclusion: The proposed AI model serves as a high-sensitivity clinical decision support tool, effectively identifying steno-occlusive lesions that may remain occult during routine radiologic interpretation. While the current specificity is affected by spurious signals, iterative refinements in post-processing—including cluster-based filtering and susceptibility artifact suppression—are expected to further optimize the diagnostic yield and streamline neurovascular workflows.